

Why is play valuable?



T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

Today

1. Play!
2. Brain-to-brain coupling during natural play
3. Overimitation

Children are noisy, variable, risk-taking,
impulsive, curious, optimistic, playful.
These are characteristic features of childhood.

Play is under siege.

25% of elementary teachers have no time for play.

79% of elementary teachers do test prep every day.

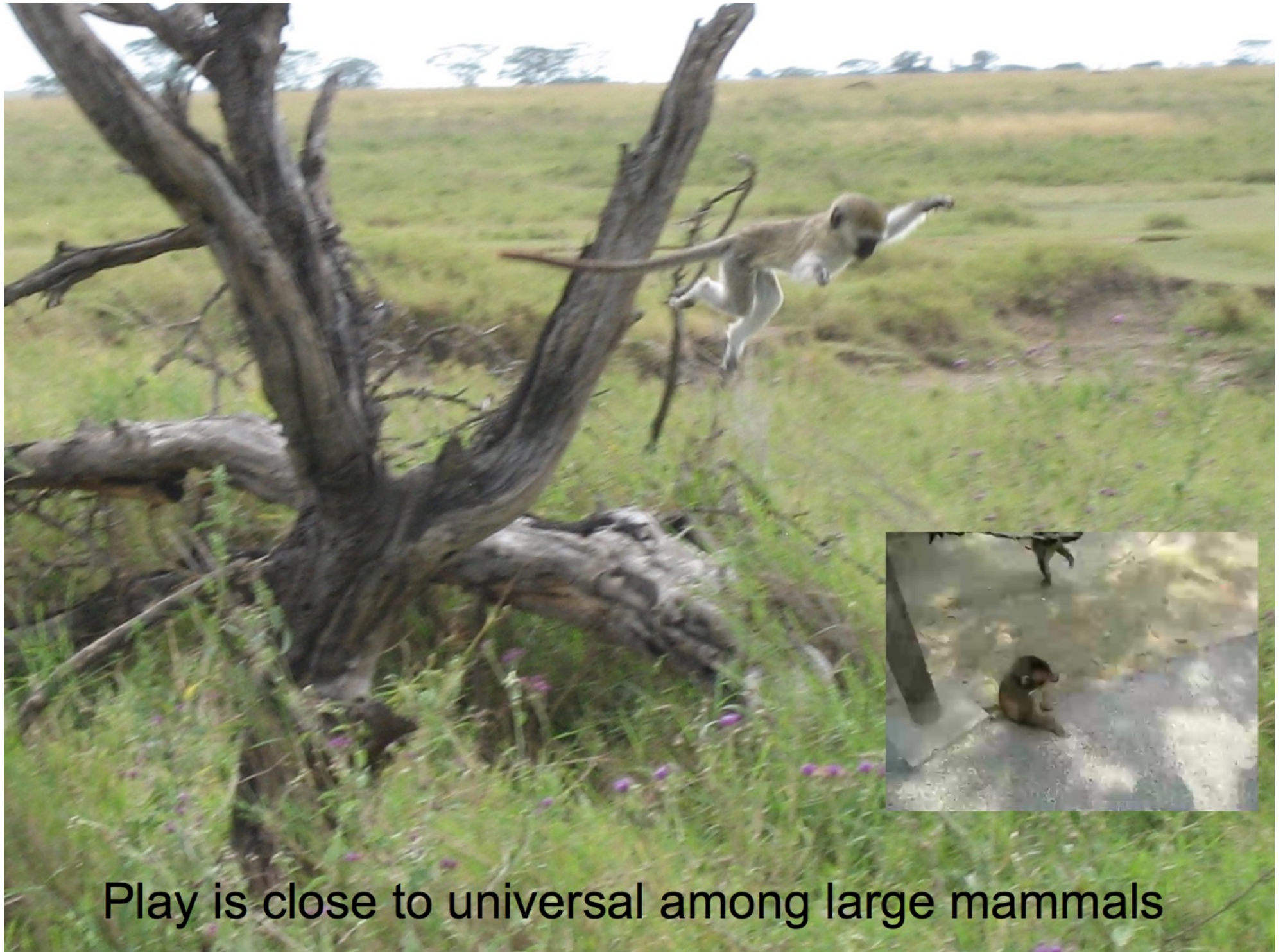
Where are the children learning?

Circle the picture.



Why is play valuable?

1. Play supports **exploration and hypothesis-testing**.
2. Play is **cognitively demanding**.
3. Play is fun, and **positive emotions** help kids learn.
4. Play promotes meaningful, **socially interactive learning**.
5. Play enables exchange of **high-quality language**.



Play is close to universal among large mammals

“Silly trunk”!



ELEPHANTVOICES

Saint Jerome on play and language learning (4th-5th century)

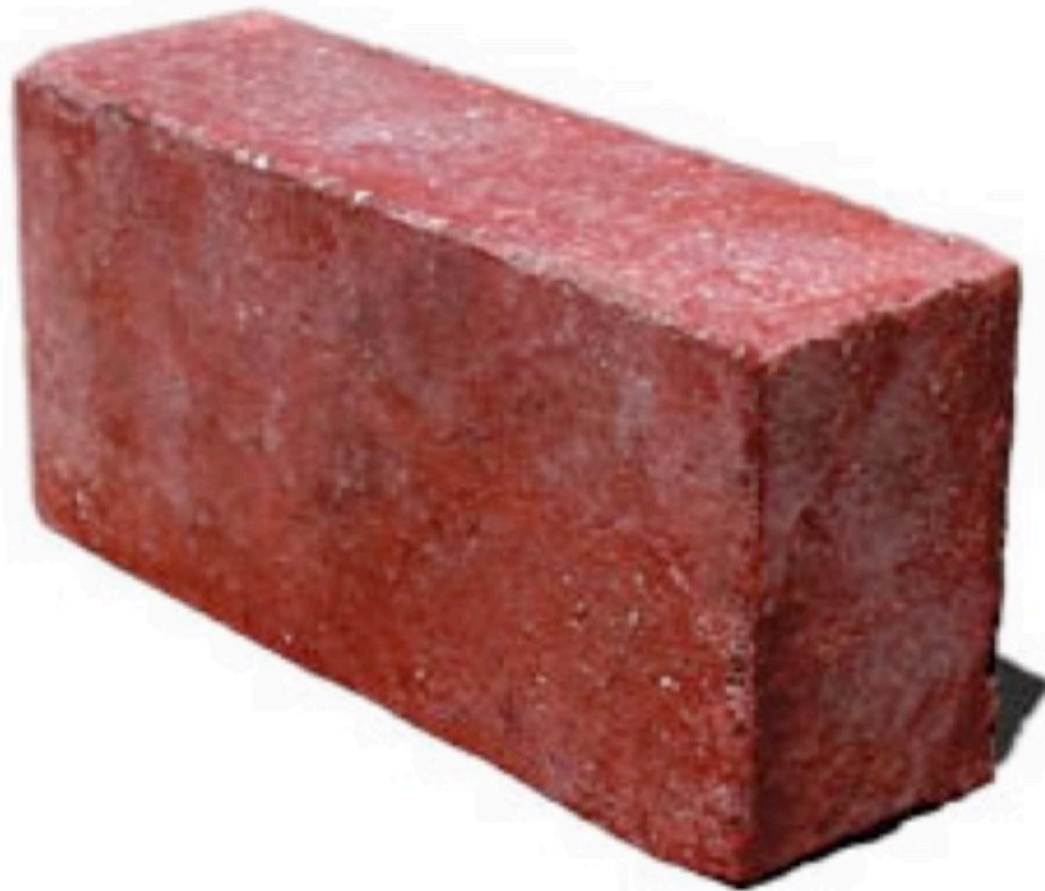


“Get for her a set of letters made of boxwood or of ivory and called each by its proper name. Let her play with these, so that even **her play may teach her something**. And not only make her grasp the right order of the letters and see that she forms their names into a rhyme, but constantly disarrange their order and put the last letters in the middle and the middle ones at the beginning that she may now them all by sight as well as by sound.”

“No behavior has proved more ill-defined, elusive, controversial and even unfashionable than play.” (E.O. Wilson)



How might you use a brick?



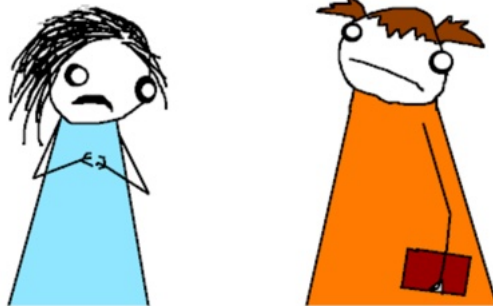
Brick Tag



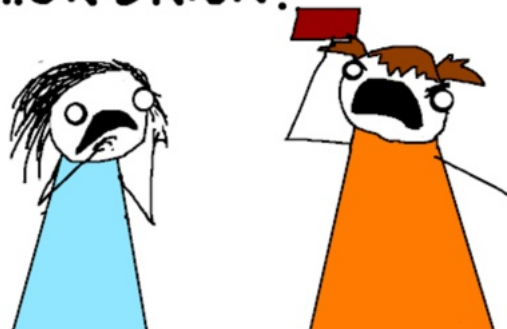
BRICK CONQUERERS!!!



TRUTH...

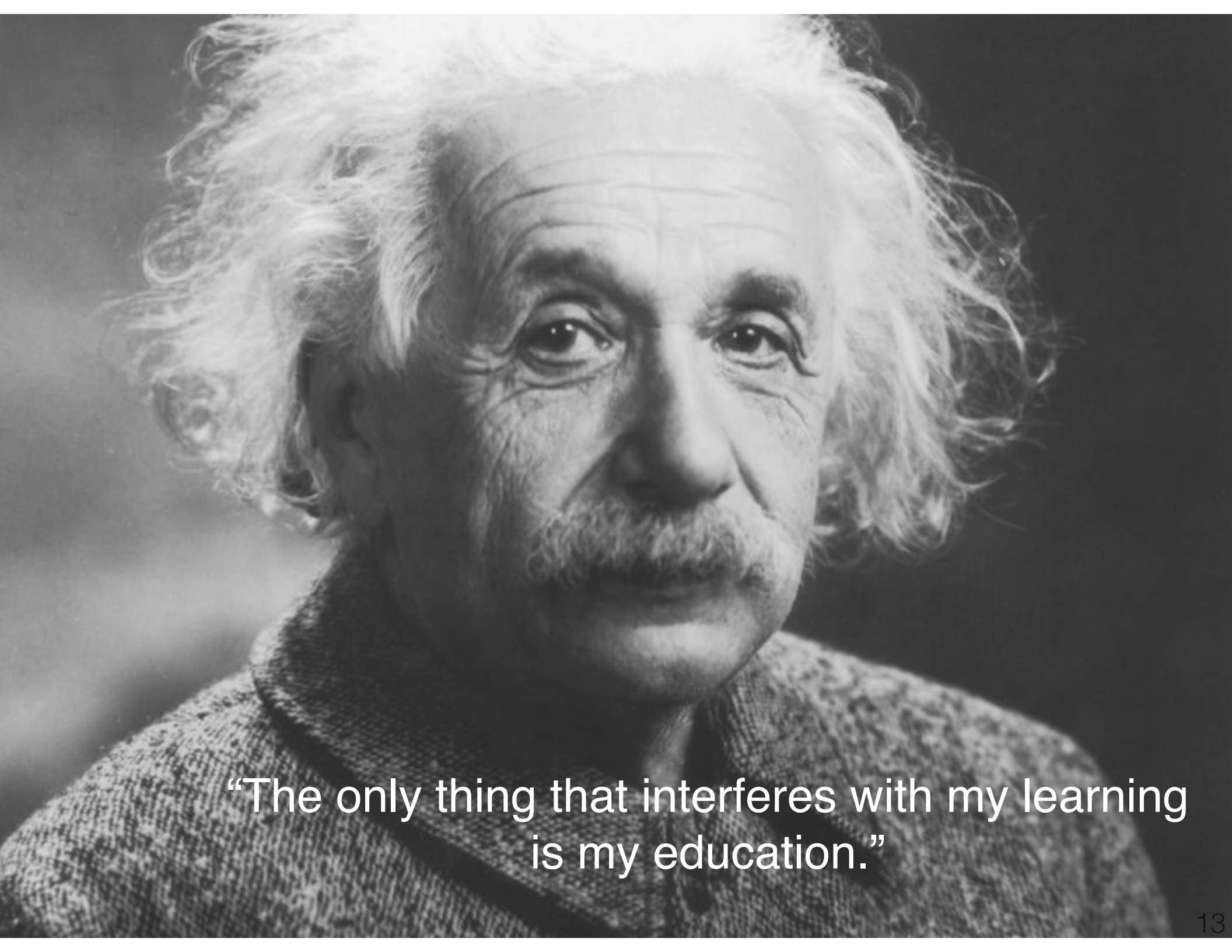


...OR BRICK!



BRICK ROULETTE





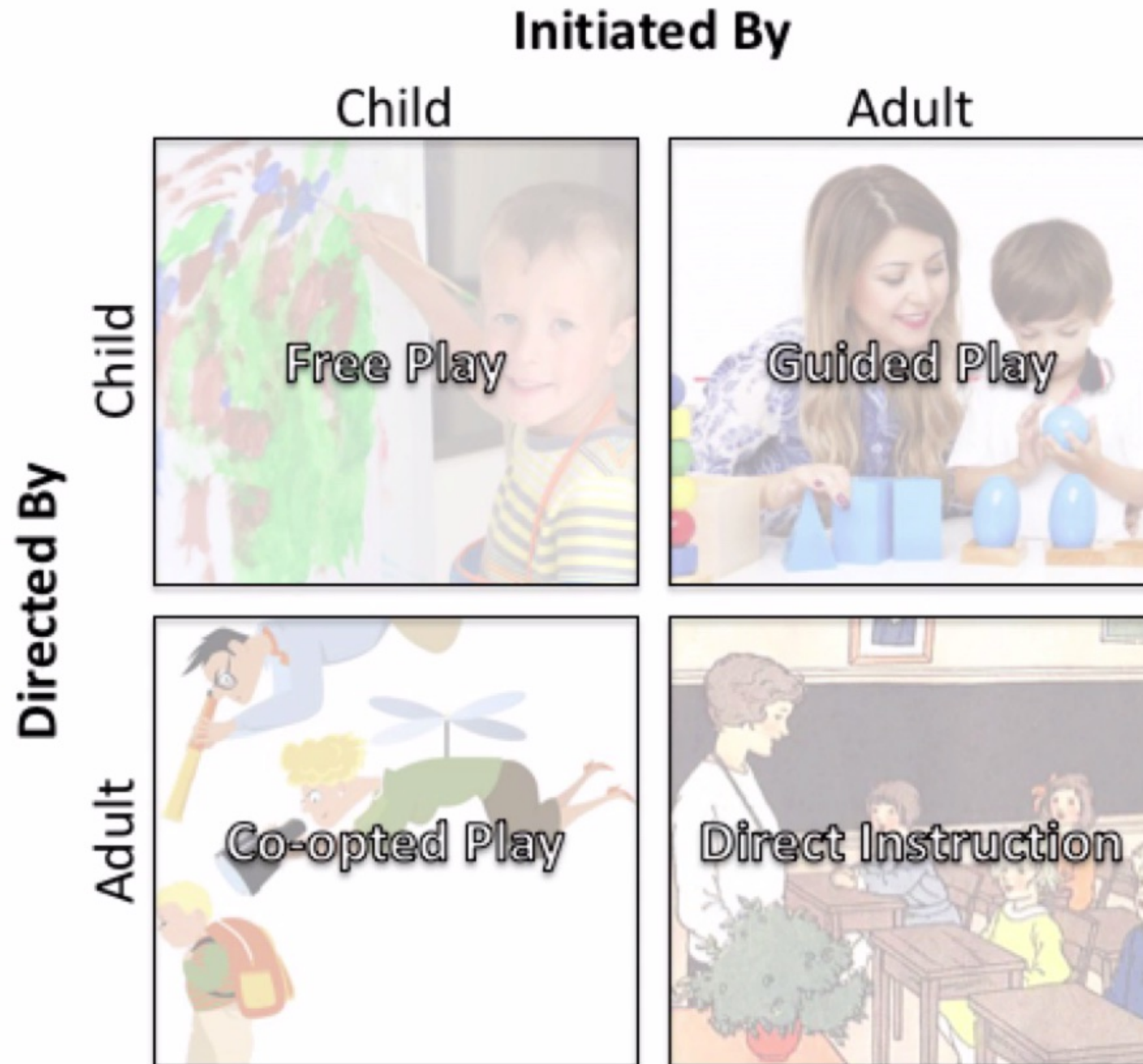
“The only thing that interferes with my learning
is my education.”

Throw out the flash cards.

The cognitive and social processes
required for imaginative play
(or for the complexities of playdates, sleepovers, etc.)
are more demanding than
sitting there looking at flash cards.

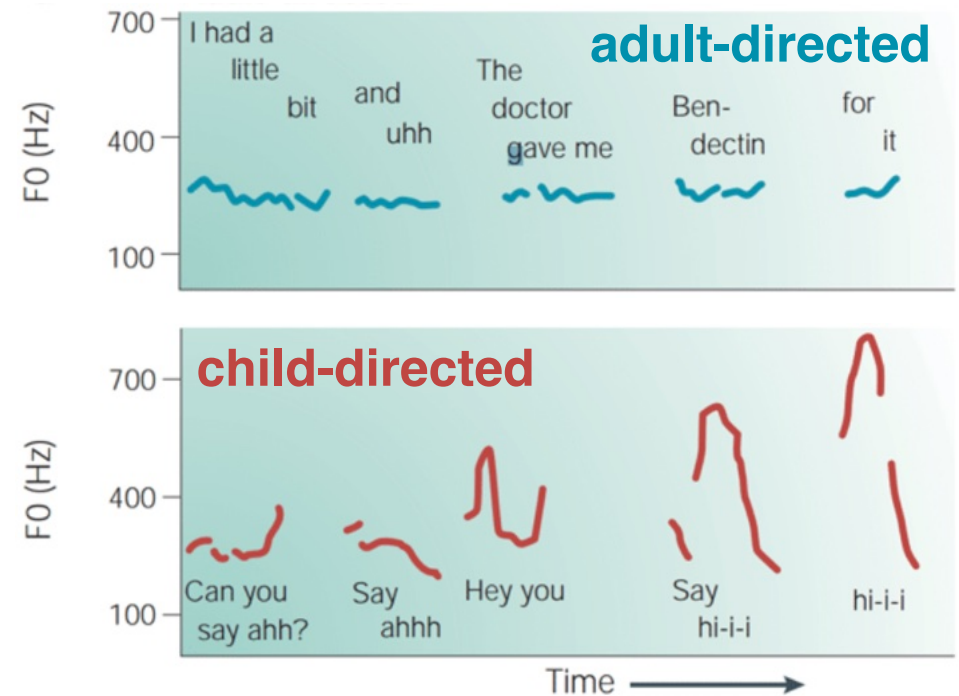
With a stick, a leaf, and a friend or family member,
great things can be achieved.

Types of play

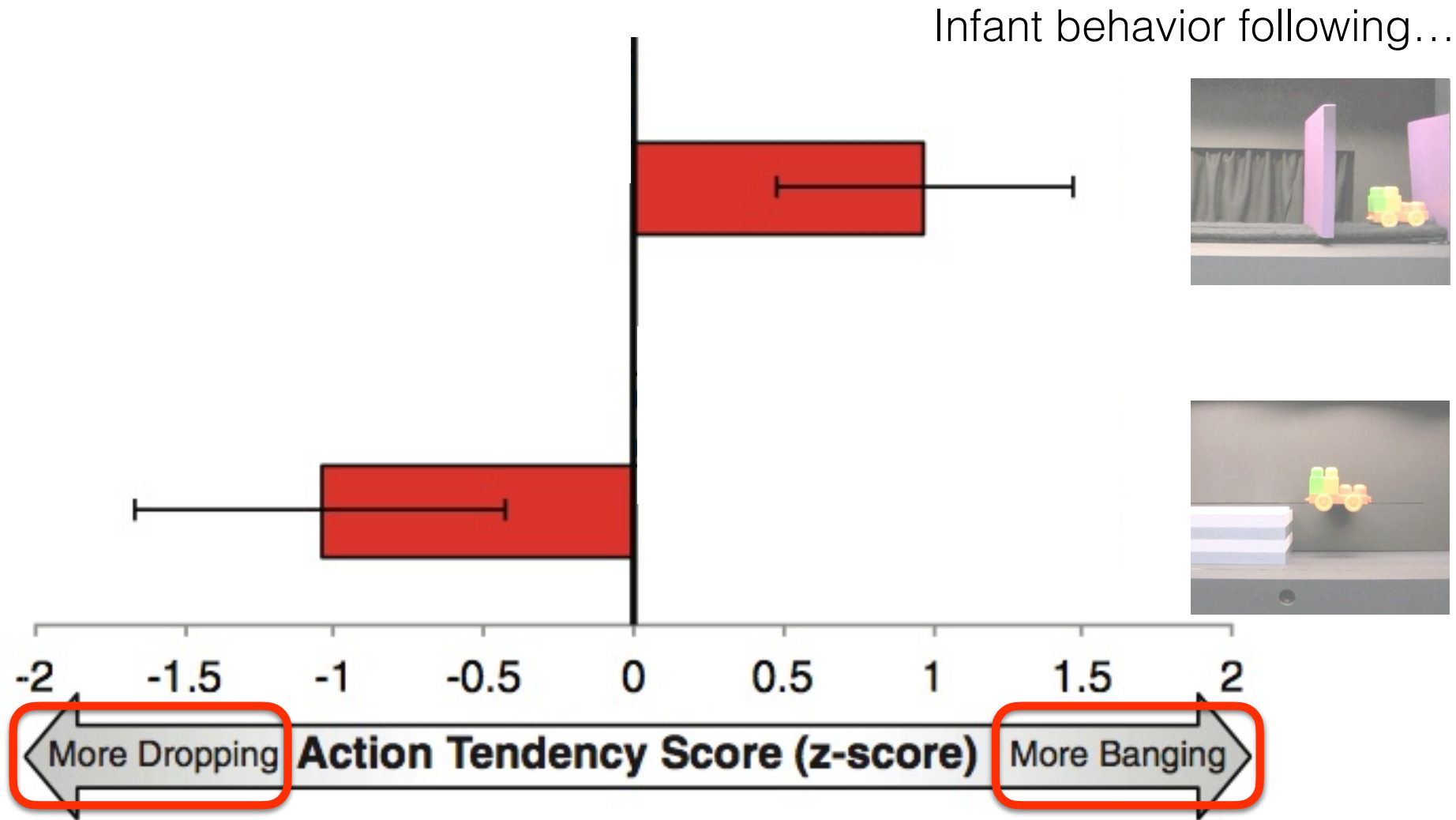


You've already learned about play

Child-directed speech is arguably playful.



You've already learned about play



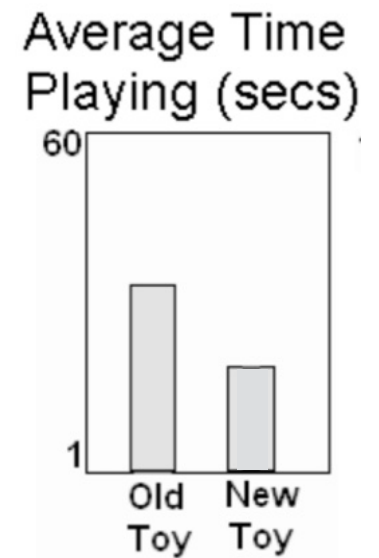
Why is play valuable?

1. Play supports **exploration and hypothesis-testing**.
2. Play is **cognitively demanding**.
3. Play is fun, and **positive emotions** help kids learn.
4. Play promotes meaningful, **socially interactive learning**.
5. Play enables exchange of **high-quality language**.

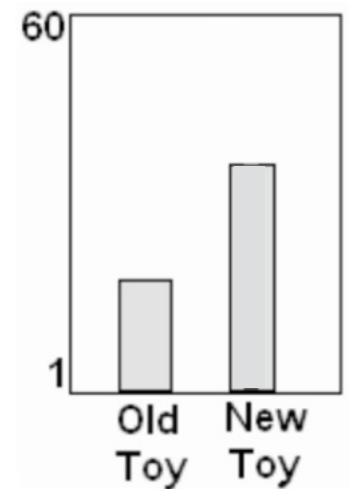
Play supports exploration and hypothesis-testing.



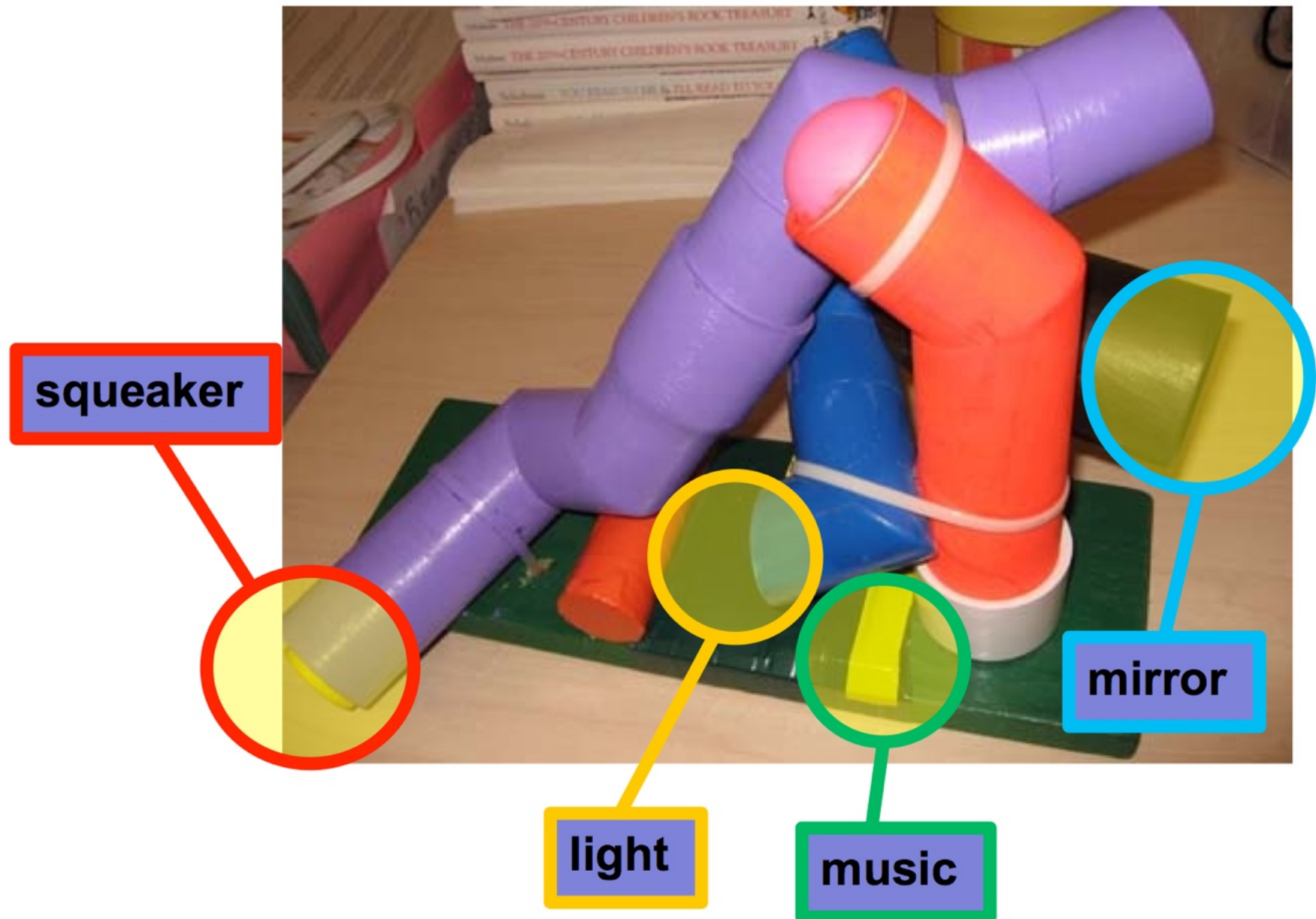
Confounded condition



Unconfounded condition

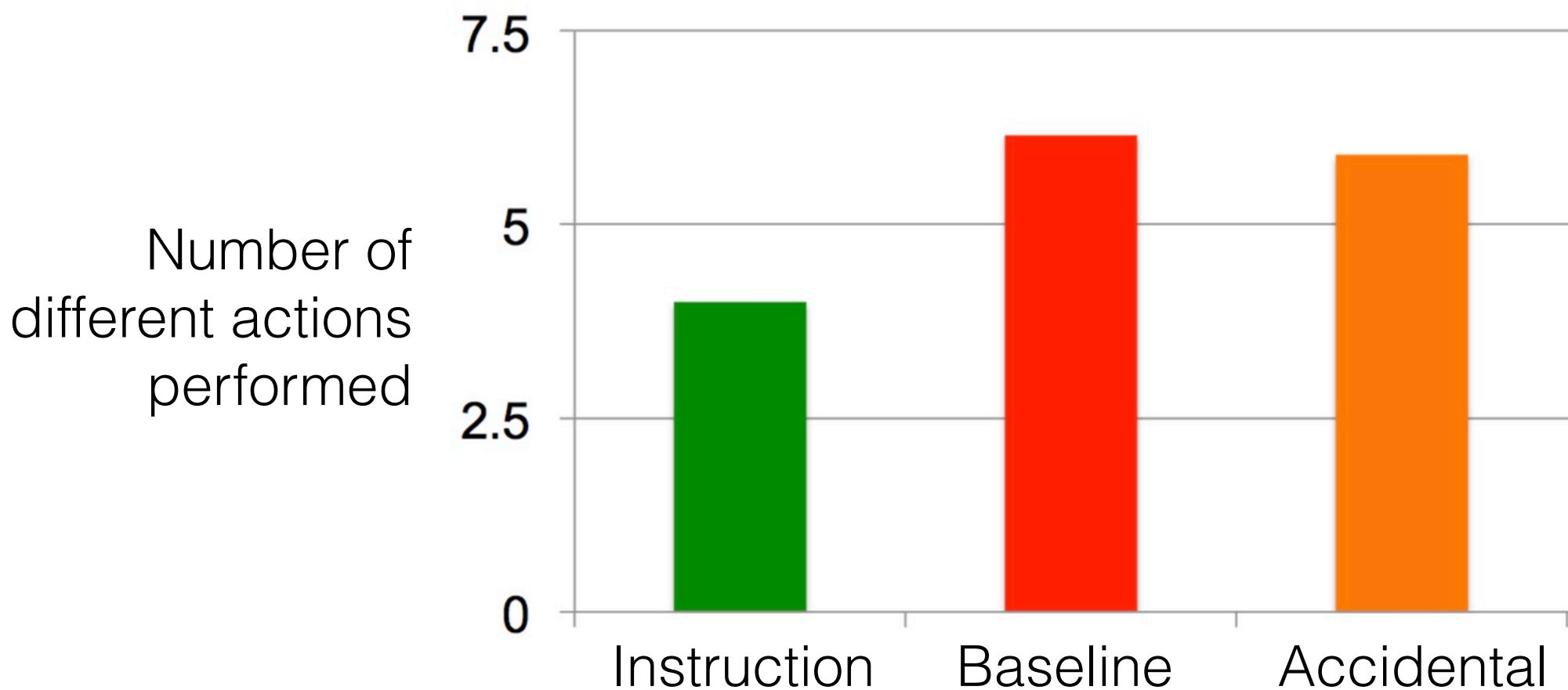


Play supports exploration and hypothesis-testing.



(Bonawitz, Shafto, Gweon, Goodman, Spelke, & Schulz, 2011; Alfieri et al., 2010)

Direct instruction limits spontaneous exploration and discovery



(Bonawitz, Shafto, Gweon, Goodman, Spelke, & Schulz, 2011; Alfieri et al., 2010)

Why is play valuable?

1. Play supports **exploration and hypothesis-testing**.
2. Play is **cognitively demanding**.
3. Play is fun, and **positive emotions** help kids learn.
4. Play promotes meaningful, **socially interactive learning**.
5. Play enables exchange of **high-quality language**.

The Batman Effect



Increased persistence (White et al., 2017)

Decreased frustration (Grenell et al., 2019)

Improved executive function (White & Carlson, 2015)

Delayed gratification (Karniol et al., 2011)

Persistence in STEM (Schachnai et al., 2022)

Play is a method for exploring
and creating social structures



calvinball



Play can also happen while reading:

Dialogic book-reading

1. Pick a book. Let the child help.

2. While reading...

- **Ask open-ended questions:** *What's happening on this page?*
- **Ask Wh-questions:** *What is the boy doing? Why is he crying?*
- **Fill in the last word:** *In the morning, the girl drinks _____.*
- **Prompt memory:** *What did the girl do after she got on the bike?*
- **Connect with child's experiences:** *We went to the doctor just like Sophia. Remember what the doctor did?*
- **Bring in new words:** *The elephant has a big nose. It's called a trunk.*
- **Prompt child to repeat new words:** *Can you say trunk?*
- **Discuss what the child said:** *That does look a bit like a horse - it's called a zebra. It has stripes, just like on your shirt, see?*

Playful learning at the grocery store



Adding playful learning to “trapped spaces” like supermarkets increased parent to child language by 33%! (Ridge, Hirsh-Pasek & Golinkoff, 2015)

Playful learning at the children's museum



Common text across conditions Welcome to the gear exhibit! Gears come in many different sizes and colors. They can be used to make many different kinds of machines.

Baseline We are interested in how parents and children spend time together at the exhibits. We would like to learn more about families visiting the exhibits at the museum. Please interact with your child as you normally would.

Encouraging explanation (explain) While your child is interacting with the gears, try asking them to explain how the gears work. Ask them to tell you about the gears or to describe what happens when they interact with the gears in different ways. Ask them questions about how the gears work or what will happen when a gear moves. Encourage them to think aloud about how the gears work.

Encouraging exploration (explore) While your child is interacting with the gears, try encouraging them to explore how the gears work. Ask them to try new things with the gears. Urge them to interact with the gears in different ways. Suggest that they figure out how the gears work, or what will happen when a gear moves. Encourage them to experiment with how the gears work.



Marlo Hampton

@2Taurean2U



Babies just be saying stuff and you just gotta nod and agree like they'll say "I like ice cream cause I don't like frogs" and you be like "damn for real?" and it helps with their development or whatever

7:29 PM · May 21, 2020 · [Twitter for iPhone](#)

74.5K Retweets **519.1K** Likes

There is SO much room for toys that are informed by developmental science.

But toys are props. They are not meant to guide the whole interaction.

Toys should be 90% child, 10% toy.

We don't need more plastic things hanging on little pegs at Target that cost \$19.99 and say 'ABC' on them and then sit in a landfill for 1 million years.

Electronic vs. traditional toys

Electronic baby toys are associated with decreases in quality and quantity of language input to babies

- fewer conversational turns
- fewer adult words
- fewer parental responses



Electronic toys do promote sustained attention, but babies don't vocalize or gesture as much (compared to traditional toys)

Traditional toys and autism spectrum disorder

Parents *and* their autistic children talk less and use less lexical diversity with electronic toys (vs. traditional toys).

A Traditional toy set



B Electronic toy set



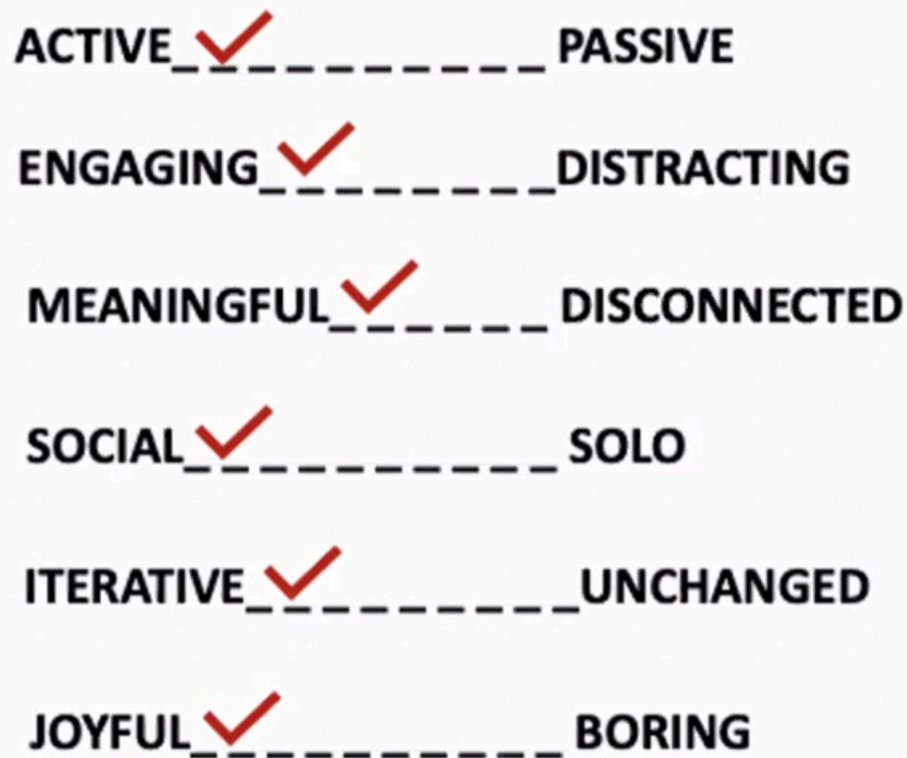
Electronic toys sometimes take over parent-child play interactions.

Main point 1:

The fast food industry fills us with empty calories, and the 'learning industry' fills us with odd notions of what is needed for successful learning and joyful lives.

Main point 2:

Play has transformative value.



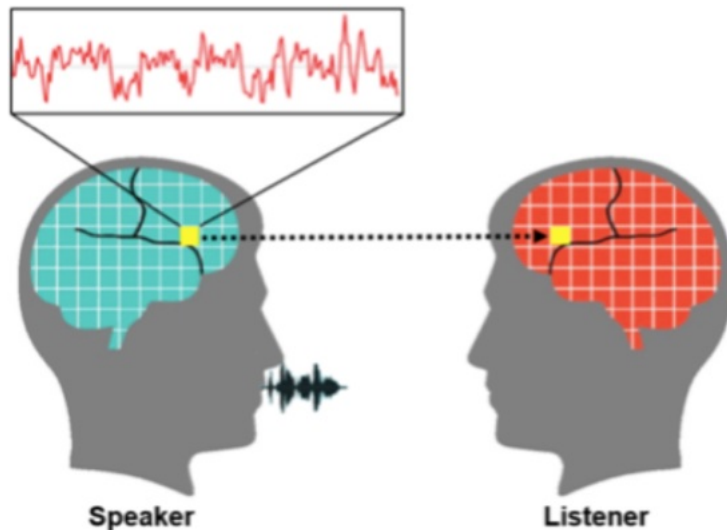
Some skepticism to keep in mind...

Has the importance of play been overblown?

Have play researchers underemphasized the utility of direct instruction and cultural transmission?

If we could learn Newton's laws (e.g., $F = ma$) just by playing in the sandbox, then we would all learn them as preschoolers.

Brain-to-brain coupling during natural play



Princeton
Baby Lab

(Piazza, Hasenfratz, Hasson, & Lew-Williams, 2020)

Together (playing)



Together (singing)

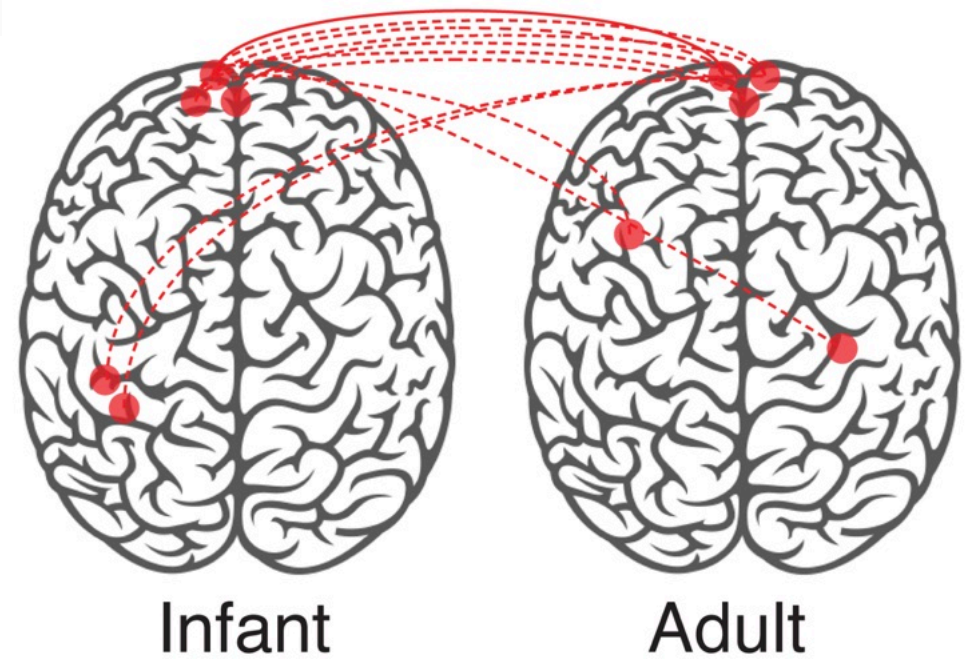
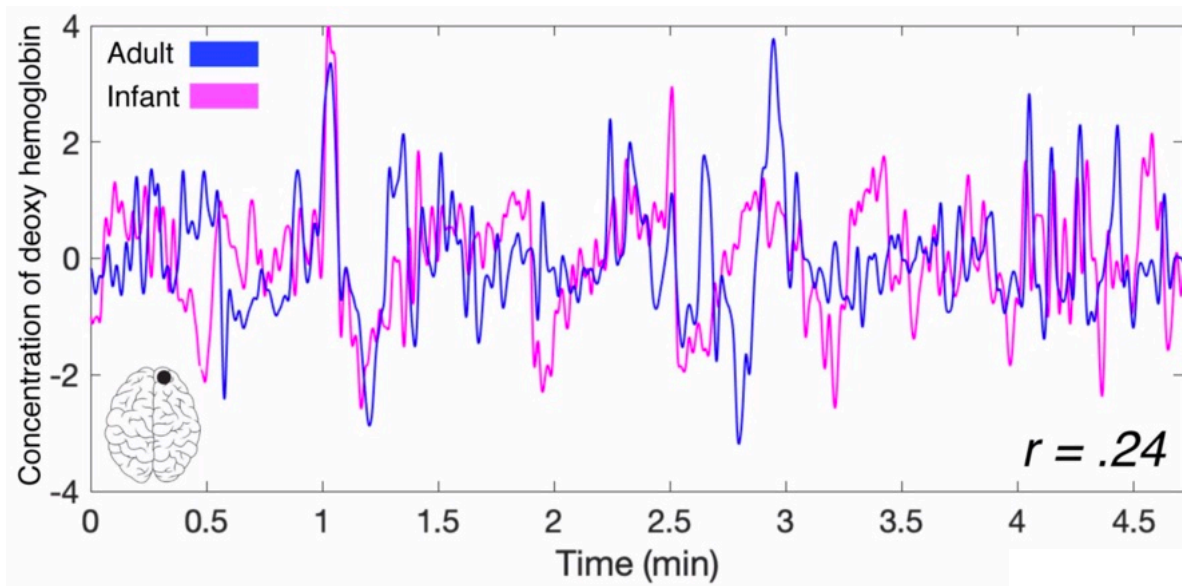


Together (reading)



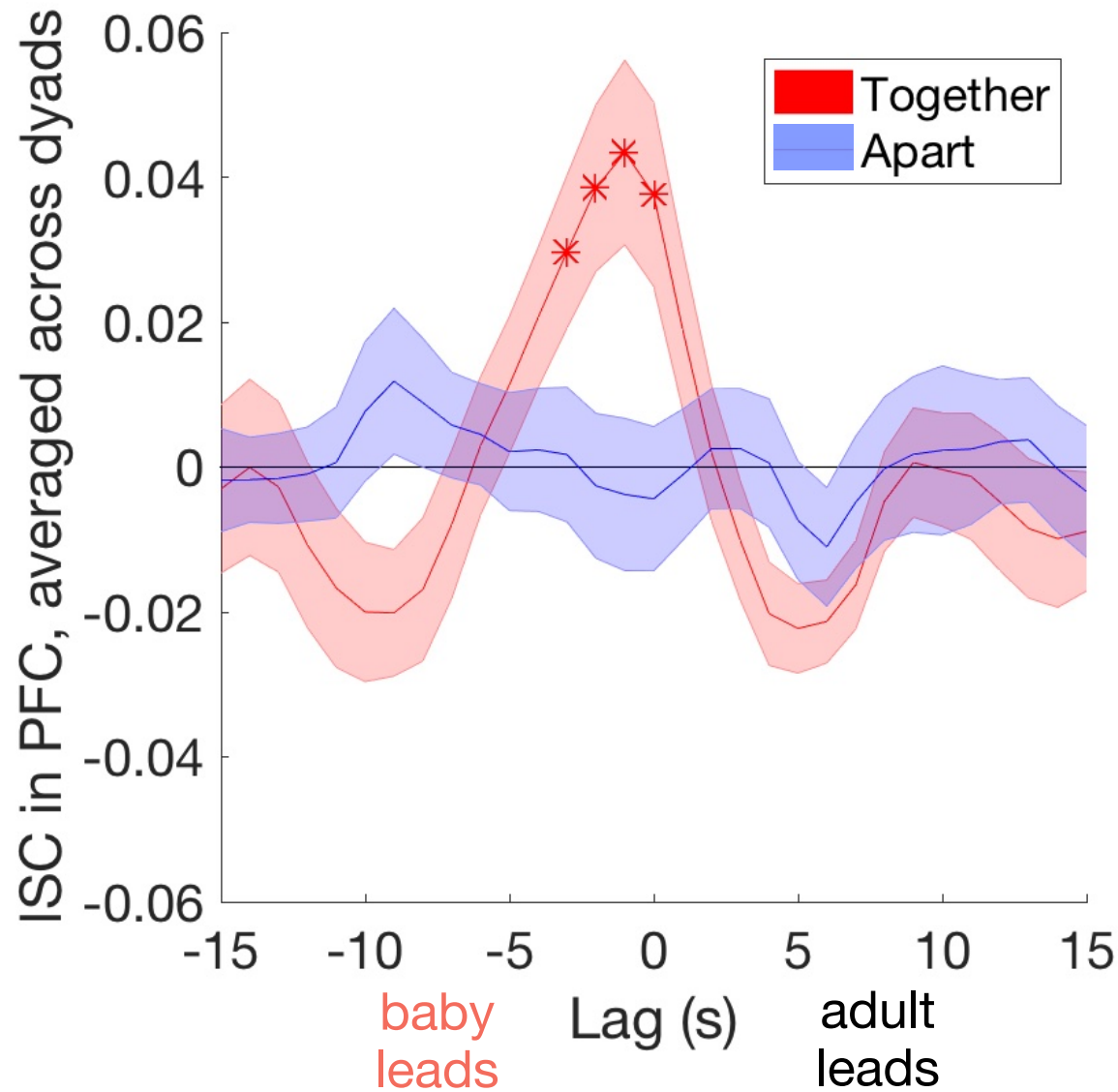
Apart





(Piazza, Hasenfratz, Hasson, & Lew-Williams, 2020)

The baby's brain activity slightly precedes the adult's brain activity!



(Piazza, Hasenfratz, Hasson, & Lew-Williams, 2020)

Adult and baby brains are coupled during natural play, and are dynamically tuned to important social cues: gaze, smiling, joint attention, infant-directed speech.

Opens new doors for studying how the developing brain organizes acoustic, linguistic, and social information during communication

This is a step toward understanding how interactions with caregivers help young children learn from the environment.

Over-imitation

Through play and interpersonal synchrony,
we imitate and over-imitate
and engage in enculturation.



<https://www.science.org/content/article/beetle-grows-termite-back-steal-food>

Links to previous lectures

Facial imitation in newborns

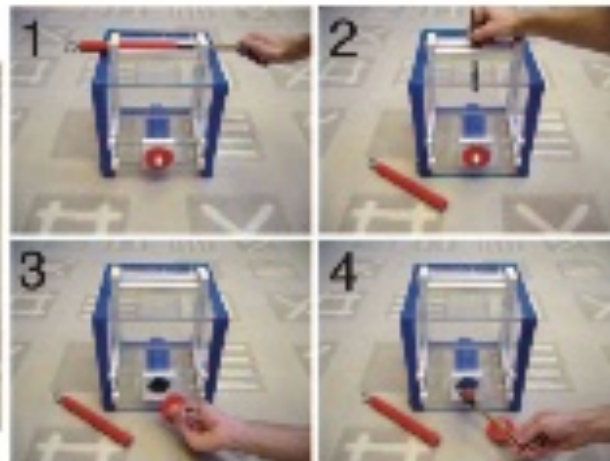


Imitation in young children

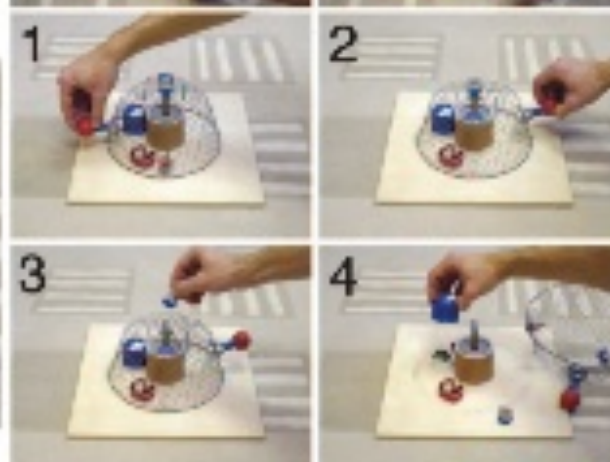
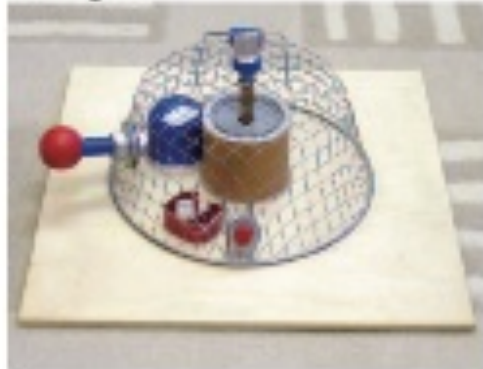


Now, an extreme example of faithful imitation of adults' actions...

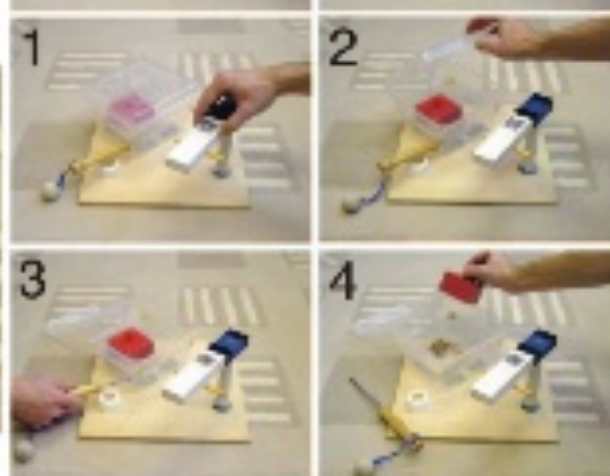
Puzzle Box



Cage



Dome



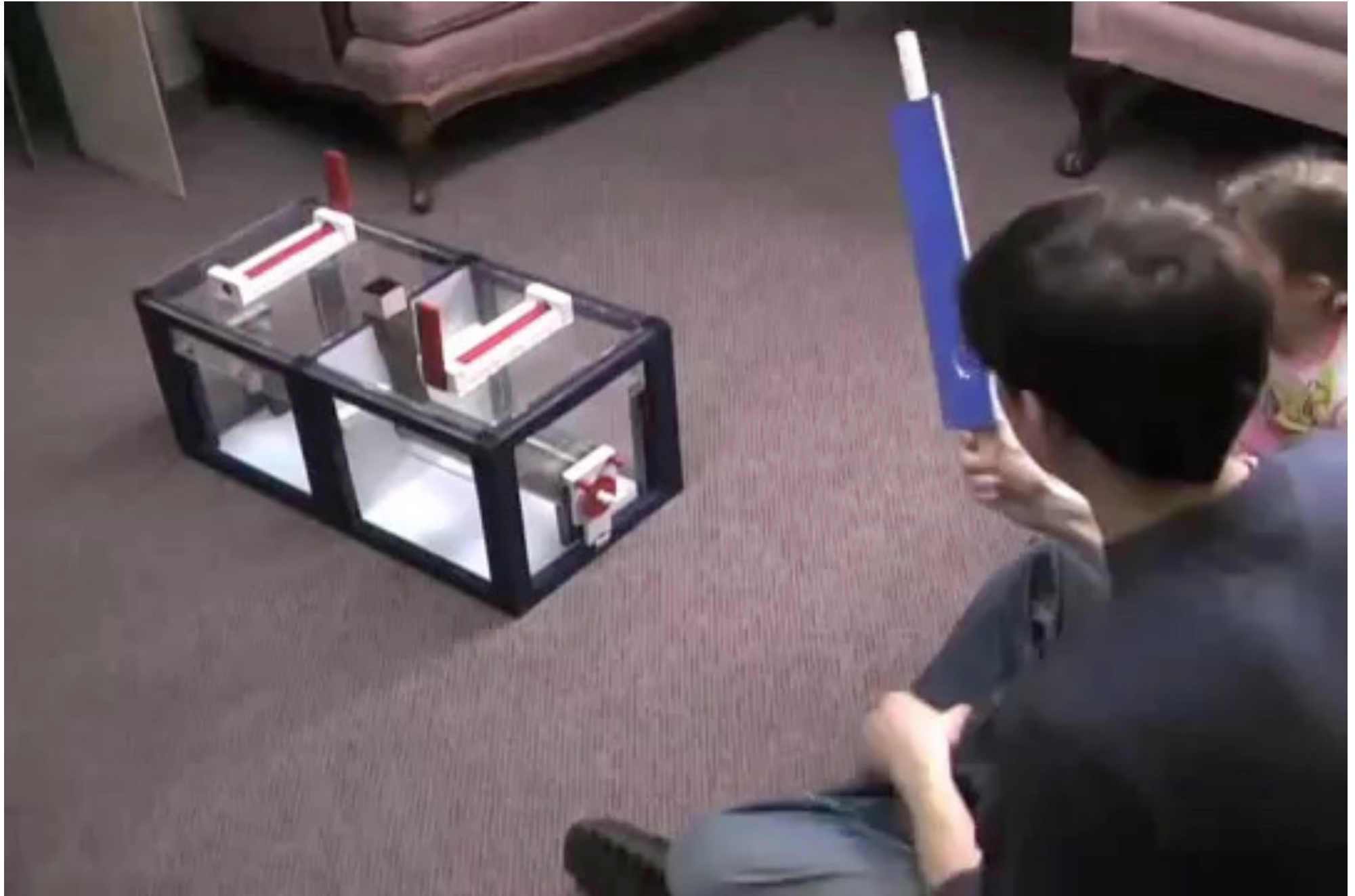
Overimitation study: Puzzle box



Overimitation study: Weird experimental context?



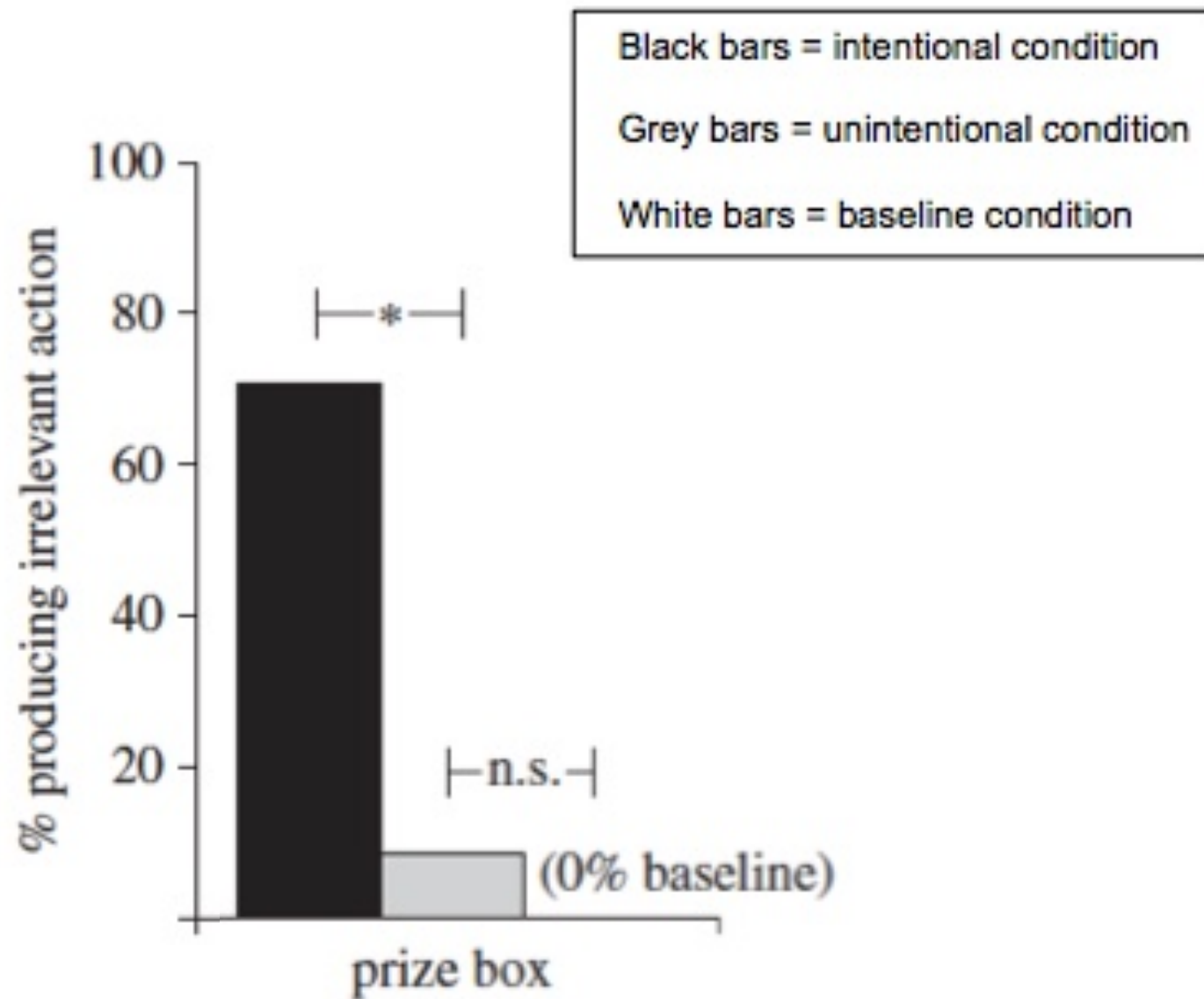
Overimitation study: Intentional vs. unintentional



Overimitation study: Intentional vs. unintentional



Overimitation study: Intentional vs. unintentional



Overimitation

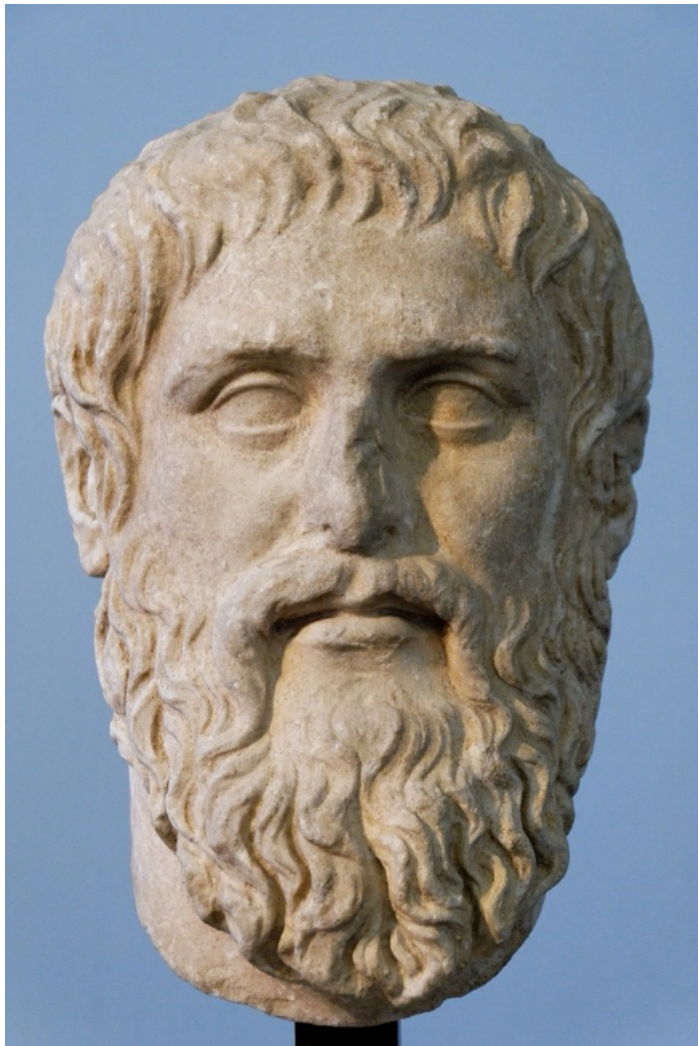
- Findings
 - Children tend to copy adults' irrelevant actions
 - Children imitate even when told to watch out for irrelevant actions, and even when speed is emphasized
 - Unintentional actions are not reproduced
- Why?
- But... we might be optimal imitators rather than over-imitators.
 - If kids see **3** people do irrelevant+relevant actions, and then see **1** person only do the relevant action, they don't imitate the irrelevant action.

Humans are able to coordinate with others in a way that other primates seemingly are not, to form a 'we' ... to create everything from a collaborative hunting party to a cultural institution.

Tomasello (2014)

Plato's Laws, Book 1

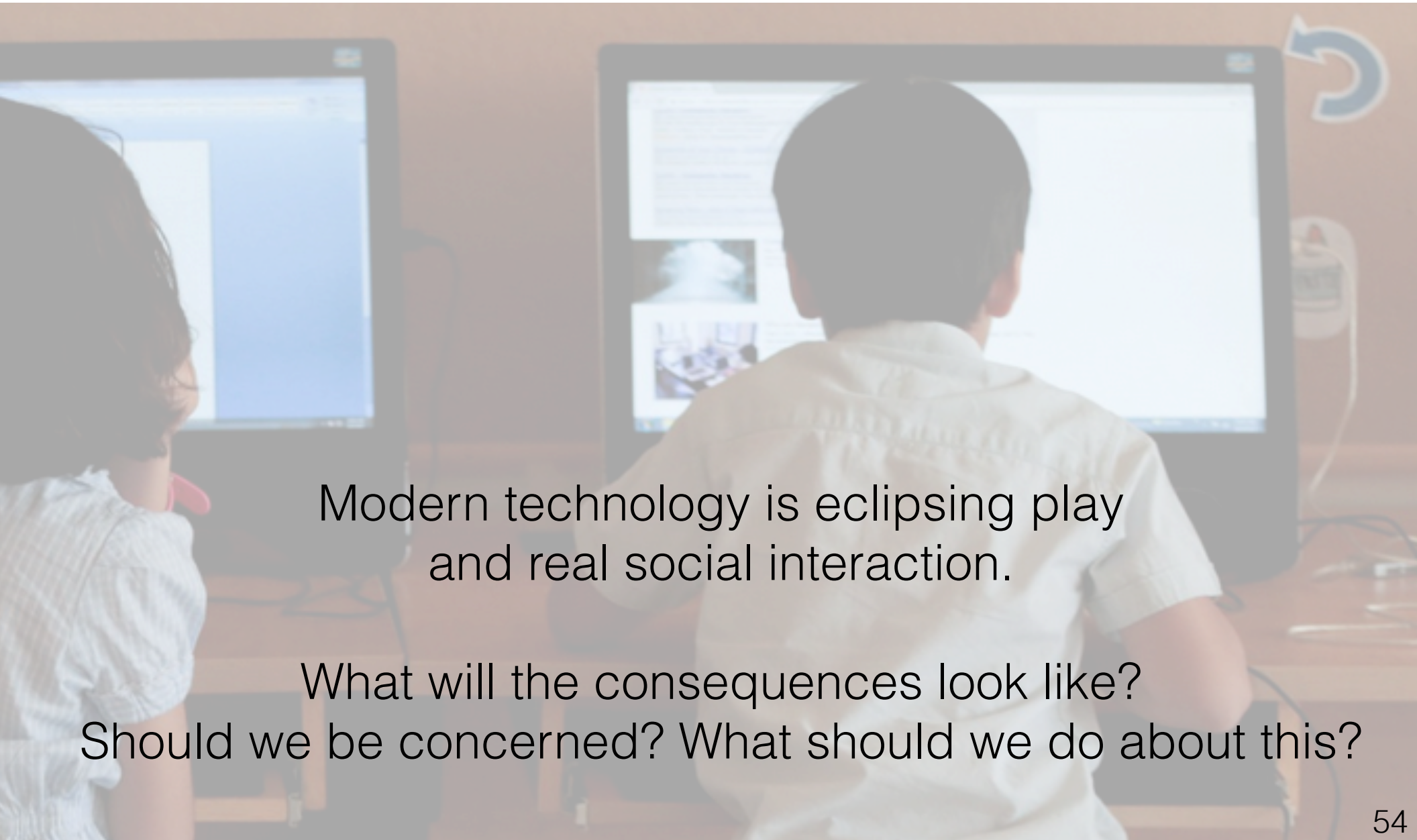
(from *Plato: Complete Works*, ed. J. Cooper)



The [person] who intends to be a good farmer must play at farming, and the [person] who is to be a good builder must spend [their] playtime building toy houses.

We should try to use the children's games to channel their pleasures and desires towards the activities in which they will have to engage when they are an adult.

The ultra-social species is now face to face with **non-social** technology.



Modern technology is eclipsing play and real social interaction.

What will the consequences look like?
Should we be concerned? What should we do about this?